2024

COMPUTER SCIENCE AND ENGINEERING

Paper : CSML-1002

(Topics in DBMS)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1** and **2** and ***any four*** questions from the rest.

1. Answer ***any five*** questions : 2×5

(a) What is a derived horizontal fragmentation? Give an example.

(b) In the context of distributed database design, why is it important to identify which part of application data are read-only items?

(c) Define the image of a database in a site for a spatially distributed database system.

(d) What is a Join graph?

(e) Explain the role of 'Block Report' in the Hadoop File System (HDFS).

(f) Why concept hierarchy is important in designing a data warehouse?

(g) How is a virtual data warehouse different from a regular data warehouse?

2. Comment on the correctness of the following statements and justify your opinion in brief (within **200** words).

Attempt ***any five*** : 4×5

(a) It is useless for a hacker to get a dump of one or more data nodes in HDFS.

(b) Dicing does not necessarily reduce the dimensions of a data cube.

(c) Primary copy locking enhances efficiency in processing the transactions in a distributed database system.

(d) Horizontal fragments are necessarily disjoint.

(e) Isolation facilitates concurrent execution of transactions.

(f) Allocation schema is location-independent even if it defines the allocation of fragments for a distributed database system.

(g) The availability of a distributed system degrades gracefully in the presence of an increasing number of faults.

3. (a) Explain the scope of vertical fragmentation towards designing of a distributed database system with a suitable example.

   (b) Explain the difference between the Basic Filter and Set Filter operations for the Hyper file data model with suitable examples of each type of filter. 4+6

4. (a) Evaluate the efficiency of using derived fragmentation for spatial databases using a suitable example.

   (b) Explain the role of local mapping schema towards integration of a heterogeneous multi-site database environment. 5+5

5. (a) Explain with an example how a single updating operation on a single data item could result in multiple updating operations on several sites when the database is physically distributed.

   (b) What is semi-join? Analyze the impact of semi-join operation on the cost of query processing in a distributed environment. 5+5

6. (a) A cube model has data in N dimensions for a particular subject orientation. The cardinalities for the four dimensions are C1, C2, C3 and C4 respectively.

      (i) Find the number of Fact and Dimension Tables in an equivalent Star Schema representation.

      (ii) State and explain the minimum number of tuples that would occur in each of the Dimension and Fact Tables in your design.

   (b) What is snow-flake schema?

   (c) Does concept hierarchy among data items on time dimension represent a completely ordered state? Explain with an example. (2+3)+1+4

7. (a) Explain the roles of Distributed Transaction Manager (DTM) and Local Transaction Manager (LTM) towards transaction management.

   (b) A 'commit' operation in database systems ensures that the changes in the database are permanent. In the 2-phase commit protocol for transaction management, two different types of 'commit' operations are performed at the coordinating site and at the participating sites. The coordinating site may decide to abort a transaction even if one or more participating sites are ready for 'local commit'. Explain if this is consistent with the ACID properties for transaction management.

   (c) How the atomicity of a transaction is ensured under 2-phase commit protocol when one of the sites fails after the coordinator issues a global commit message? 4+3+3

8. *Tourismo* is a travel agency that books hotel rooms for its clients in big cities across India. The information to be stored is given below :

   Destination information : location id, place name, nature of place (sea-side, mountains, big cities, etc.)

   Hotel information : hotel id, location id, hotel name, number of rooms, room charges

   Guest information : guest id, name, address, phone number

   Occupancy : hotel id, guest id, room number, check-in date, check-out date

Design a distributed database solution for hotel room reservation system to be accessed from multiple offices of *Tourismo* in multiple cities considering the following frequent queries/operations :

Request for reservation of room : If a guest wants to book multiple rooms in same or more than one hotel for different period, separate reservations are to be done for each stay bed and period of booking.

Availability of room : the system is to check for availability of room in a particular hotel for a period.

Revenue query : the system is to be used to enquire about the revenue generated in a single hotel or, in a particular location or, in the entire chain for a period.

Your design should include the definition of global schema, fragmentation schema, and allocation schema. State your assumptions clearly that would justify your design. 10